IDC MarketScape

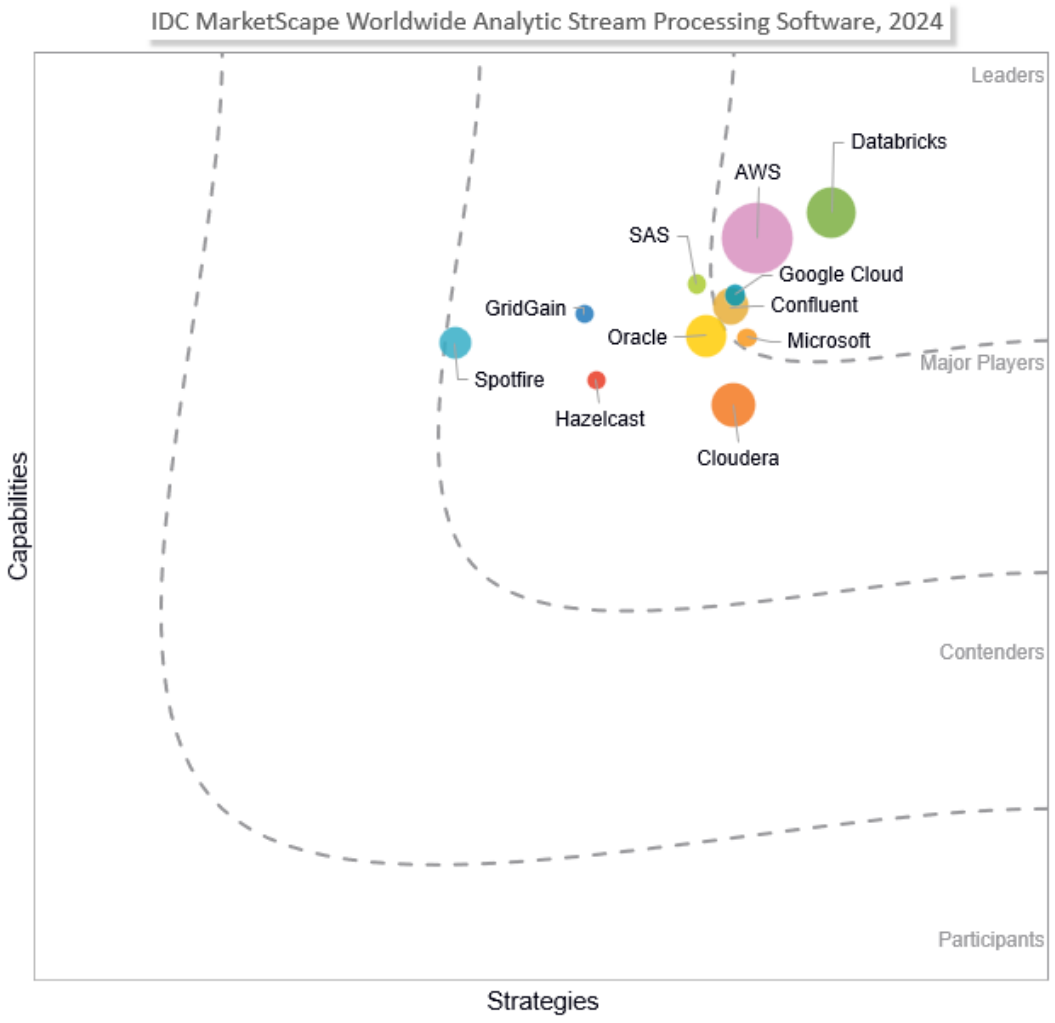
IDC MarketScape: Worldwide Analytic Stream Processing Software 2024 Vendor Assessment

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IDC MARKETSCAPE FIGURE

FIGURE 1

IDC MarketScape Worldwide Analytic Stream Processing Software Vendor Assessment



Source: IDC, 2023

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Please see the Appendix for detailed methodology, market definition, and scoring criteria.

IDC OPINION

The speed of business has increased, as organizations need to respond to and make decisions based on what is happening now, not what happened yesterday, last week, or last month. The number of real-time analytic solutions in organizations has been increasing, as so too has the amount of money being spent in the streaming analytics software markets. The worldwide event stream processing market as measured by IDC has seen a 21% compound annual growth rate (CAGR) for the five-year period from 2017 to 2022 and a forecast 30% CAGR for the five-year period from 2022 to 2027. Streaming solutions are present in all major geographies around the world, across all major industries, but they are more prevalent within financial services. Popular use cases include financial analysis, real-time recommendations, providing service guarantees, cybersecurity, risk management, media and gaming, Internet of Things (IoT) applications, geospatial analysis, geofencing, and vehicle tracking.

Analytic stream processing software is available as a service or as client-managed implementations on global hyperscale cloud services and from independent software vendors. Not all analytic stream processing solutions are the same because there is not just one way to process events in a stream in real time:

- Processing events directly in the streaming data technologies (event brokers)
- Processing events at the source, such as capturing database change events
- Ingestion of events into high-performance in-memory data stores for processing
- Ingestion of events into a data lake (house) for processing

Each method has varying degrees of latency that are supported from nominal (less than 200ms) to ultra-real time (less than 40ms). Events can be stateful or stateless; are either pushed, pulled, or persistent; and one event at a time or in a micro batch.

Regardless of how the events are processed, the common denominator across all is time, more specifically, time windows. Analytics starts with querying data, and stream analytics requires a time boundary or window for the query. Windows can be fixed, sliding, tumbling, session based, custom defined, memory optimized, and time-stamp controlled. This represents the biggest change that a data or application engineer needs to understand when working with streaming data compared with working with at-rest data.

Analytics include event aggregation, filtering, transformation, routing, and enrichment for use in analytical models. Output from real-time streaming analytics can go into real-time dashboards, decisioning algorithms, applications, data lakes, data lakehouses, data warehouses, back into the stream, time-series databases, or artificial intelligence (AI) models. As the event brokering technologies, event types, and formats have evolved over the years, so too has the technology being used to process events.

There is a general tension in the market between proprietary stream processing engines and open source engines. Solutions using proprietary engines express performance, scalability, and direct integration with sources and targets of analytic stream processing. Open source solutions started with Apache Storm, then Apache Spark became more prominent, and now Apache Flink is becoming prominent in stream processing solutions. There are also innovations in the open source community in

support of standard data formats such as Avro, JSON, and Parquet and stream programming frameworks such as Apache Beam. Most popular programming languages are supported, with SQL, Java, and Python being most prominent.

We are entering an era of AI everywhere, and the top 4 business drivers for AI as measured by IDC require real-time streaming data: improvements in quality, business agility, employee productivity, and customer experience. Analytic stream processing will become critical for success in AI. The need for real-time data is prevalent in top AI use cases by industry as measured by IDC:

- 3 of the top 4 AI use cases in the federal and local governments require real-time data
- 5 out of the top 6 AI use cases in banking and manufacturing require real-time data
- 4 out of the top 5 AI use cases in retail and utilities require real-time data

From automation of business processes to automated management of technology in the enterprise, real-time data is critical across all these use cases because business happens in real time. Batch processes were introduced because of technological limitations. The ability to process data in near real time has been available for at least two decades, but only now are we starting to see organizations transition as technology gets better at responding to what is happening in real time and trust in that technology increases.

Every vendor evaluated in this IDC MarketScape provides real-time analytic event processing and integration with AI technologies to meet the demands of running a digital business in modern times, where the speed of business is increasing and the time for organizations to respond to changes is decreasing. This IDC MarketScape evaluates vendors across stream processing capabilities and go-to-market strategies based on qualitative and quantitative market research that provided insight into what was important for customers wanting to compete as digital businesses.

IDC MARKETSCAPE VENDOR INCLUSION CRITERIA

The vendor inclusion list for this document was designed to accurately depict the vendors that are most representative of any given analytic stream processing software buyer's selection list. Vendors were then surveyed and further investigated to ensure that the offerings qualified with both capabilities and strategies related to the cloud integration market.

Critical to this research effort was for the vendor to meet the inclusion criteria. Any vendor participating in this IDC MarketScape had to showcase that it met the following:

- Offers analytic stream processing capabilities as part of a sustainable business, whether offered standalone or part of a larger technology portfolio (if part of a larger portfolio, the analytic stream processing solution must be technology agnostic — there must be no requirement to have adopted a solution from that portfolio, for example, a software-as-a-service [SaaS] application from that vendor])
- Offers analytic stream processing that fully supports analytical use cases including data ingestion and sharing, event aggregation, transformation, and filtering and can perform stream integration
- Has at least $25 million in revenue for 2022, specifically from analytic stream processing software/services offering regardless of where it is deployed
- Operates in North America and recognizes a minimum 5% of revenue from at least two additional regions

- Goes to market horizontally across industries and across horizontal processes of an enterprise

ADVICE FOR TECHNOLOGY BUYERS

Every solution acquisition initiative needs to start with the requirements for the solution. Streaming analytics can be a solution specific to one project, but it is better leveraged as a shared enterprise service capable of being used across multiple use cases and projects. As such, buyers should be considering enterprise-level requirements while looking at project-level requirements. Some requirements to consider include the following:

- What type of latency do I require for my analytic stream processing use cases —nominal (less than 200ms), real time (greater than 40ms, less than 200ms), or ultra-real time (less than 40ms)?
- Is the use of open source important? Latency requirements, how events are processed, and integration with analytics solutions will influence your decision.
- The data lake, lakehouse, and warehouse technologies being used in the enterprise may influence the decision as each solution has different alignments in the ecosystems that exist in the market.
- What types of event broker technologies are being used across the enterprise? Kafka is supported by all and most of the commonly used message-oriented middleware software. Pulsar support is growing.
- While many of the analytic stream processing software have graphical programming environments with low-code and no-code capabilities, most also support the use of commonly used programming languages. Assess programming skills within your organization against the languages supported by these technologies.
- Support of industry-specific capabilities, content, models, and so forth is sporadic among the vendor solutions evaluated in this IDC MarketScape. If having industry-specific capabilities is important, choose wisely.
- What levels of security and privacy need to be supported by the solution? Do you have FIPS or FedRamp requirements, HIPPA, GDPR, or SOC requirements?
- Governance and control are key elements of data processing in a digital business. Consider what types and level of intelligence you require about events and event processing. Catalogs, schema registries, lineage, and quality enforcement are available within the market.

Working with streaming data requires a mind and paradigm shift for developers and operations personnel; identify the levels of training you will require for developers and operations staff. Some of the open source stream processing engines can be very complex to manage, which is why most of the vendors that offer open source support make their offering available as a managed service. Consider whether a managed service is required, or will your IT organization manage the solution? What level of monitoring and alerting is required, and does it need to integrate with existing systems management investments. What deployment options are preferred, such as cloud, on premises, or hybrid?

All vendors in this evaluation offer flexibility in licensing and pricing but to varying degrees. Consider the most optimal approach to licensing for your organization based on your use cases, technology, and budget constraints.

The evaluation of analytic stream processing software within this IDC MarketScape is based on the criteria that IDC created, informed by research. Buyers need to perform their own evaluation against

their own requirements within the constraints of their own IT, procurement, and industry environmental constraints. Every evaluation process should also include proof of concept or what is often referred to as a "bake-off." These evaluations of the software need to be done against data used by your organization, in the context of your organizational structures, and within the constraints of your organization's technology environment.

Once the selection process has completed, the hard work of implementation begins. Change management will be key to successful implementation because use of stream analytics will change the way people work and how the enterprise functions.

VENDOR SUMMARY PROFILES

This section briefly explains IDC's key observations resulting in a vendor's position in the IDC MarketScape. While every vendor is evaluated against each of the criteria outlined in the Appendix, the description here provides a summary of each vendor's strengths and challenges.

AWS

After a thorough evaluation of AWS' strategies and capabilities, IDC has positioned the company in the Leaders category in this IDC MarketScape for worldwide analytic stream processing software.

Amazon Web Services (AWS) is a worldwide hyperscale cloud service provider, was founded in 2006, and is headquartered in Seattle, Washington, United States. AWS has several data-oriented services available in its cloud, including its Managed Service for Apache Flink (previously known as Amazon Kinesis Data Analytics). The Amazon Managed Service for Apache Flink makes it easy to build and run real-time stream processing applications using Apache Flink. Amazon Managed Service for Apache Flink provisions and configures Flink clusters and orchestrates Flink job management, allowing the developer to focus on higher-value functionality in streaming data pipelines. The Amazon Managed Service for Apache Flink has several companion products including Amazon Glue Streaming ETL, Amazon Glue Schema Registry, Amazon Glue Data Catalog, and Amazon EMR.

Quick facts about Amazon Web Services include:

- **Product name:** Amazon Managed Service for Apache Flink
- **Employees:** 100,000+ (estimated)
- **Global footprint:** Operates across all global and within many regional locations around the globe
- **Top industry areas:** Financial services, manufacturing, telco, media and entertainment, and high tech
- **Cloud:** Available on the AWS Cloud
- **Pricing model:** Usage based by time duration and level of compute
- **Partner ecosystem:** Global partner ecosystem through the AWS Partner Network
- **Interesting fact:** AWS has offered a Managed Apache Flink service since 2018 (formerly called Kinesis Data Analytics) and supports tens of thousands of applications.

AWS was one of the first companies to offer a managed Flink service, and because of this, the company has depth of knowledge and capabilities in operationalization of analytic stream processing using Flink. The Amazon Managed Service for Apache Flink has proven availability, security, and

durability of streaming solutions using the Apache Flink open source project. The Amazon Managed Service for Apache Flink is serverless, it supports all Flink APIs, and customers can use Java, Python, Scala, and SQL to create their applications. It supports Flink's exactly once processing semantics and state management configs, and it has over 40 prebuilt connectors and supports over 200+ custom connectors. The Amazon Managed Service for Apache Flink Studio is an interactive experience for stream querying and promoting queries into long-running Flink apps. AWS supports customers' access to specialist Flink engineering, solution architects, and support, including Flink PMC members and committers.

AI-assisted development is an active and growing opportunity for data stream processing that AWS is pursuing, anchored by Amazon Q, which was announced at AWS re:Invent in the fall of 2023. Amazon Q is a generative AI-powered solution that provides opportunities for automation of processes from requirements gathering through development, deployment, and ongoing management of solutions running on the AWS Cloud.

Strengths

- **Compatibility and language support.** The Amazon Managed Service for Apache Flink supports multiple types of event brokers such as Kafka, Pulsar, RabbitMQ, and MQTT. It also supports all popular programming languages including C++, C#, Java, Python, SQL, Scala, Ruby, Julia, Go, and R. It also supports the use of Apache Beam and Spark.
- **Market presence and partner ecosystem.** The Amazon Managed Service for Apache Flink is part of the AWS Cloud, which offers availability across all global regions, providing sales and service support in 53 countries. AWS has a vast partner network inclusive of channels, services, and technology companies. Early in 2023, AWS launched a partner program explicitly focused on the Amazon Managed Service for Apache Flink.

Challenges

- **Awareness.** The Amazon Managed Service for Apache Flink was formerly named Kinesis Data Analytics. The market did not know that Flink was behind the Kinesis Data Analytics service, and AWS was missing the opportunity as Flink grew in popularity for stream processing. This is the premise behind the recent rebranding into the Amazon Managed Service for Apache Flink. Apache Flink requires a steep learning curve to build and operate, and AWS saw an opportunity to be more transparent and better align with growing industry trends.
- **Cloud deployment alternatives.** The Amazon Managed Service for Apache Flink is only available on the AWS Cloud. This, however, does not preclude it from being able to connect to event brokering technologies running in hybrid and multicloud environments using secure connections.

Consider AWS When

Consider AWS when your organization has a disposition toward AWS as a cloud service provider and, as such, much of your streaming data infrastructure is also contained within the AWS Cloud. You may also consider AWS if you are using Flink as a stream processing engine and want to leverage a managed service to accelerate the learning curve and/or reduce complexity of operations, managing Flink clusters, OSS patches, and maintenance.

Cloudera

After a thorough evaluation of Cloudera's strategies and capabilities, IDC has positioned the company in the Major Players category in this IDC MarketScape for worldwide analytic stream processing software.

Cloudera is a private company headquartered in Santa Clara, California, United States. Cloudera was founded in 2008, providing commercial support of the open source Hadoop project, and was a significant force in driving the big data era forward. At the end of 2018, Cloudera and Hortonworks (an alternative Hadoop solution) merged into the new Cloudera company, merging their platforms and software together into one solution now known as the Cloudera Data Platform. Cloudera remains committed to the open source community as many of its solutions are based on open source projects. Cloudera has been trying to shed its legacy image of "the Hadoop company," now that Hadoop as a data technology has taken a back seat to modern alternatives. SQL Stream Builder and its use of Apache Flink is a great example of how Cloudera has moved beyond its Hadoop roots and continuing to embrace open source software to deliver solutions in the market. Stream processing solutions built with Cloudera give organizations the power to respond to critical events at the speed of their data while streamlining development and deployment with an efficient and nimble architecture that can scale to meet all streaming needs.

Quick facts about Cloudera include:

- **Product name:** Cloudera Data Platform, SQL Stream Builder, and Apache Flink
- **Employees:** 3,500 (estimated)
- **Global footprint:** Deployed in 71 countries across 9 global regions
- **Top industry areas:** Financial services, telecommunications, and utilities
- **Cloud:** Available as-a-service on Amazon Web Services, Microsoft Azure, Google Cloud, and IBM Cloud
- **Pricing model:** Based on number of Cloudera Compute Units
- **Partner ecosystem:** Multiple partners in all major global regions
- **Interesting fact:** A multinational oil and gas corporation was able to realize a 90% faster response time to cybersecurity threats with a Cloudera-based log analytics solution, leveraging NiFi, Kafka, and Flink as part of the Cloudera Data Platform.

Cloudera's tenure in the big data software market provides a depth of experience and capabilities in enterprise-scale solutions, delivering enterprise-level qualities of service in modern data environments. Having to transform itself from a Hadoop-based infrastructure to modern data technologies, Cloudera has learned what it means to pivot and adapt, offering its experience and knowledge to customers and prospects that face similar challenges. Cloudera solidified its use of Flink for stream processing in 2020 when the company acquired Eventador. NiFi and Flink are tightly coupled to allow for integrated security and governance but with enough autonomy that each component can be modified individually. Cloudera is committed to leveraging open source software for its data lake and stream processing capabilities, inclusive of Apache NiFi, Kafka, Flink, Iceberg, and related projects.

Strengths

- **Stream processing accessibility.** SQL Stream Builder provides a SQL-based user interface for building stream processing applications, making stream processing accessible to more user personas through its SQL interface, while also being code-friendly for data engineers. Its

native integration with schema registry and stream lineage capabilities assists with pipeline life-cycle management, troubleshooting, and impact analysis.

- **Market presence.** Cloudera has a good market presence and a healthy level of investment in product development, and the company has fostered a community around its brand through strong support of open source software projects. Cloudera's implementation of Apache NiFi for data movement, Apache Kafka for event brokering, and Flink for event processing supports connections to over 450 different technologies, with MiNiFi extending NiFi capabilities to the edge.

Challenges

- **Containerization.** Executable software in the Cloudera Data Platform supports running on Kubernetes containers, virtual machines, and bare metal, but not all of Cloudera's streaming products currently support Kubernetes. Cloudera's road map is focused on modernizing its streaming platform into microservices that run in Kubernetes containers.
- **Market awareness.** While Cloudera's market presence is a strength, much of that has been built up through its big data legacy, and many of its stream processing customers represent an expansion within the current customer base. Cloudera is still trying to shake its Hadoop stereotype, making the market more aware of its capabilities in modern data technologies, inclusive of event brokering and processing in operational and analytical use cases, to attract new logos.

Consider Cloudera When

Consider Cloudera when your organization is confident in and prefers the use of modern open source data technologies in a multicloud and hybrid environment and requires enterprise-class scalability, reliability, and security. Cloudera is a good alternative for organizations that would benefit from a no-code/low-code stream processing development environment, which also has extensive connectivity from the edge to the cloud. If use of generative AI is on your road map, Cloudera has some innovations on its road map that will support real-time retrieval-augmented generation (RAG) processes, and it is developing a Hugging Face ReadyFlow interface to retrieve and load test data sets into the data lakehouse.

Confluent

After a thorough evaluation of Confluent's strategies and capabilities, IDC has positioned the company in the Leaders category in this IDC MarketScape for worldwide analytic stream processing software.

Confluent was founded in 2014 by the original creators of Apache Kafka, a project that was initially created at LinkedIn. Confluent is a publicly traded company headquartered in Mountain View, California, United States. Confluent is committed to helping organizations transform into a new paradigm for data management, one of event-driven and real-time data in motion. Kafka was initially created for event brokering and operational applications, and it has evolved into being used for analytic stream processing. Confluent Cloud offers event brokering in Kafka and analytic stream processing as managed cloud-native software as a service that is elastic, resilient, and performant. With the acquisition of Immerok, Confluent released a fully managed Confluent Cloud version for Apache Flink later that year, which will also likely drive further usage of event stream processing. Confluent Cloud removes the complexities of implementing and operating Kafka infrastructure, connectors, processing, and governance, allowing customers to focus on building streaming data pipelines and real-time applications to achieve operational and strategic business outcomes.

Quick facts about Confluent include:

- **Product name:** Confluent Cloud
- **Employees:** Over 2,750
- **Global footprint:** Available in all global regions
- **Top industry areas:** Financial services, retail, and high tech
- **Cloud:** Available on Amazon Web Services, Microsoft Azure, Google Cloud, Alibaba, Oracle Cloud Infrastructure, and the IBM Cloud
- **Pricing model:** Pay as you go, based on time duration, and level of compute
- **Partner ecosystem:** Global partner ecosystem
- **Interesting fact:** A North American space agency is taking Confluent Cloud beyond the clouds and into space. The agency is leveraging Confluent Cloud to create the General Coordinates Network (GCN), which is a public collaboration platform for the astronomy research community to share alerts and rapid communications about high-energy, multi-messenger, and transient phenomena. Observatories all over the world can publish alerts in real time to the network when they witness any phenomenon in the sky. Anyone can subscribe to those alerts, whether it is someone at the agency or a researcher at an academic university or an individual with a telescope.

As a data streaming platform, Confluent Cloud seeks to serve the needs of a large pool of potential customers through a broad coverage approach. In keeping with Confluent's general-purpose approach, Confluent Cloud supports publish and subscribe, point to point, and request-response, covering all primary messaging patterns as well as supporting almost all common programming languages except R. Confluent Cloud offers users many event processing capabilities that provide a strong foundation for a general-purpose model, such as event routing, event transformation, event filtering, load balancing, and event replay. While lacking topic hierarchies, Confluent Cloud allows users to do advanced content-based and dynamic routing.

Confluent Cloud is highly infused with predictive AI, interpretive AI, and generative AI, which can provide for a wide range of potential use cases within each of these three categories. For predictive AI, Confluent Cloud enables users to undertake fraud detection, supply chain management, MLOps feature selection, model training, inventory, and so forth. Confluent Cloud's interpretive AI use cases involve chatbot enhancement, content classification, compliance monitoring, reasoning agent automation, and so forth. Finally, for generative AI use cases, Confluent Cloud can add content to chatbot responses, generate code, execute semantic searches, and generate content. Confluent Cloud's broad AI use case potential further reinforces Confluent's aim of being a general-purpose platform.

Strengths

- **Stream governance.** Confluent has built an extensive set of stream intelligence capabilities that enable stream governance. The company supports event discovery through its stream catalog and data portal, runtime schema validation provided by a schema registry, stream lineage capability to trace data flowing through the stream, and the ability to tag sensitive event data flowing within the stream. These capabilities contribute to improving the quality of data in the stream, stream tracing, and compliance with internal and external policies and regulations.
- **Partnering program and multicloud support.** Confluent has a large and diverse set of global channels, implementation, and technology partners. Confluent is available on all public clouds,

including Amazon Web Services, Azure, GCP, Alibaba, Oracle Cloud Infrastructure, and the IBM Cloud.

Challenges

- **Event broker support.** Confluent Cloud's implementation of ksqlDB and Kafka Streams only supports use of Kafka as an event broker. Confluent has introduced a new managed service for Apache Flink that may provide support for additional brokering technologies.
- **Market focus.** Confluent is focused on converting prospects that run the open source edition of Apache Kafka into Confluent Cloud customers. Although this does provide Confluent with a large opportunity, the company could benefit from an expanded focus to attract new logos into the Kafka ecosystem.

Consider Confluent When

Confluent's origins with Kafka make it a popular choice for customers with investments in Kafka and Flink (with its new managed service for Flink offering) in multicloud and hybrid environments. It is also suitable for organizations that are wanting to move toward a real-time data environment but do not have Kafka or Flink management skills. Confluent is also a good choice for organizations with developers skilled in SQL because ksqlDB and the forthcoming managed Flink service support this popular data engineering language.

Databricks

After a thorough evaluation of Databricks' strategies and capabilities, IDC has positioned the company in the Leaders category in this IDC MarketScape for worldwide analytic stream processing software.

Databricks was founded in 2013 by researchers and engineers at the University of California, Berkeley, who played a key role in the development of Apache Spark. Databricks has become a prominent player in the big data and analytics space, offering a platform that facilitates collaboration and accelerates time to insight for organizations dealing with large and complex data sets. Databricks offers a unified analytics platform that seamlessly integrates real-time data processing with the lakehouse paradigm, machine learning (ML), and AI. The Databricks approach to real-time data processing enables customers with a range of skill sets, from data analysts and scientists to software developers, to easily derive insights from streaming data at scale. Databricks has recently adopted *data intelligence* as a term to define its platform, with Unity Catalog being the hub around which all data ingestion, processing, analytics, data science, and stream processing are instantiated and executed. Data intelligence within the Databricks platform includes analyzing usage patterns to manage partitions and execution for optimal query and processing performance and cost.

Quick facts about Databricks include:

- **Product name:** Databricks Data Intelligence Platform
- **Employees**: 6,000+
- **Global footprint:** Operates in all global regions
- **Top industry areas:** Financial services, retail, and telco
- **Cloud:** Available on Amazon Web Services, Microsoft Azure, Google Cloud, and Alibaba
- **Pricing model:** Pay as you go based on time duration and level of compute utilized
- **Partner ecosystem:** Global partner ecosystem

- **Interesting fact:** Databricks coined the term *data lakehouse* in 2020, when Ali Ghodsi, cofounder and CEO, used the term in a keynote to the dismay of the marketing team, which had not approved the use of the made-up word. Databricks now claims that 74% of global enterprises have implemented a lakehouse architecture.

Databricks is an open platform with unified batch and stream processing engines, with APIs built on open source technologies including Apache Spark, Delta Lake, MLflow, Redash, and Delta Sharing. Databricks unifies stream processing with the data warehouse and data lake in one lakehouse platform. Customers can build end-to-end real-time data streaming, real-time business intelligence and SQL analytics, and real-time AI use cases in one platform. Unity catalog provides customers with one common security and governance model for historical and real-time data within the lakehouse.

Databricks is investing heavily in AI capabilities, including its acquisition of Mosaic ML in 2023, enabling organizations with the ability to securely build their own foundational models, with their own proprietary data, at more attractive price points. Databricks has also demonstrated use of generative AI in its Lakehouse IQ technology, which when prompted in natural language, can build an entire analytical pipeline using data contained within the lakehouse.

Strengths

- **Developer experience.** Low-code, no-code, code-friendly, and code-only alternatives make stream processing available to multiple personas. It also supports Git integration to enable CI/CD testing and deployment to production environments. Unity Catalog provides a comprehensive set of capabilities for event discovery, schema registry, stream lineage, and sensitive data tagging, so that developers have the full context of data being processed in the stream.
- **Go-to-market focus.** Databricks is focusing its efforts on operational expansion and customer success, and efforts are evident through improvements being made in partnering programs and customer success programs, and Databricks has a healthy level of investment in research and development. Databricks' success can be seen in the positive market momentum it has experienced in the past five years.

Challenges

- **Streaming engine support.** As the originator of Apache Spark, Databricks fully supports Spark streaming, but that is the only streaming engine it supports. Spark streaming has some technical limitations that prohibit its ability to process data at an ultra-real-time latency (less than 40ms).
- **Market awareness.** Databricks was known before the era of stream processing and because of its original focus in batch processing, customers and prospects sometimes overlook the maturity, breadth, and depth of Databricks' streaming capabilities.

Consider Databricks When

Consider Databricks when your organization is wanting to unify streaming and batch workloads for both analytics and AI in one platform while also taking control of processes and data by enabling governance with data intelligence. Your organization prefers cloud computing and wants flexibility to select the cloud of choice because Databricks is available on Amazon Web Services, Azure, Google Cloud Platform, and Alibaba.

Google Cloud

After a thorough evaluation of Google Cloud's strategies and capabilities, IDC has positioned the company in the Leaders category in this IDC MarketScape for worldwide analytic stream processing software.

Google is a hyperscale cloud service provider that was founded in 1998 and operates in all global regions except for China. Google is headquartered in Mountain View, California, United States. The Google Dataflow service is a serverless data processing platform built on the Apache Beam model providing a unified approach for historical (batch) and real-time (streaming) data processing. Dataflow is accessible with a wide variety of developer SDKs and no-code Dataflow Templates and supports over 60 sources and sinks to handle historical and real-time data processing requirements.

Quick facts about Google Cloud include:

- **Product name:** Dataflow
- **Employees:** 190,000+
- **Global footprint:** Available in over 200 countries
- **Top industry areas:** High tech, media and entertainment, retail, telecommunications, financial services, and healthcare
- **Cloud:** Available on Google Cloud only
- **Pricing model:** Pay per use by time and duration of compute
- **Partner ecosystem:** Global partner ecosystem
- **Interesting fact:** Dataflow has been built and proven at scale, leveraging real-world learnings and innovations from not only a wide variety of customers in many industries but also from Google's own use of the stream processing stack for billion-user products such as Search, Ads, YouTube, Maps, and Photos.

Google Cloud's mission is to help every organization accelerate its ability to digitally transform its business and industry with an intelligent, unified, and open data and AI ecosystem. Google Cloud Dataflow, built on the Apache Beam model, has a unified approach for batch and streaming data processing, enabling organizations to train data engineers on a single data processing and programming model, providing efficiencies in training and retaining skill sets through the organization. Google Cloud is bringing together all its data and analytics capabilities under the BigQuery umbrella, catering to customer requests for simplicity in packaging and pricing, providing a single stock-keeping unit (SKU) for the suite from which customers can pick individual capabilities without needing to buy new products or services.

Google Cloud enables real-time machine learning with Dataflow ML, which builds on ML-specific capabilities built into the Apache Beam model and accompanied by availability of GPU resources within the Google Cloud. Google is adding Gemini (formerly known as Duet AI) to BigQuery, enabling generative AI-powered capabilities to improve user productivity through assisted capabilities for onboarding, operations, troubleshooting, and smart suggestions.

Strengths

- **Compatibility and interoperability.** Apache Beam enables batch and stream processing in Spark and/or Flink. Dataflow also supports multiple event brokers in addition to the Google

Cloud Pub/Sub technology solution. Dataflow supports the most popular programming languages used in data engineering, and it also supports Swift and Typescript.

- **Focus on growth.** Google is focused on growth, with a dedication to its partner program, innovation, and road map communication and a high rate of on-time delivery of road map items. Google's dedication to customer success was echoed in feedback received from customer references that spoke with IDC.

Challenges

- **Industry-specific support.** Dataflow Templates are provided for a wide variety of horizontal deployment patterns, available for any industry. While none are specific to any one industry, Google does provide guidance on industry-specific applications of the horizontal templates.
- **Vendor lock-in.** Google Dataflow is only available on the Google Cloud Platform, connecting to event brokering technologies implemented in the Google Cloud or to hybrid data sources via secure networking.

Consider Google Cloud When

Consider Google Cloud when your organization is looking for a single unified platform for stream and batch processing and/or has a disposition toward Google Cloud as a cloud service provider, with much of your streaming data infrastructure contained within the Google Cloud Platform. The use of Apache Beam provides efficiency in terms of developer productivity and processing flexibility, helping reduce learning curves associated with stream processing.

GridGain

After a thorough evaluation of GridGain's strategies and capabilities, IDC has positioned the company in the Major Players category in this IDC MarketScape for worldwide analytic stream processing software.

GridGain is a private company headquartered in Foster City, California, United States. GridGain was founded in 2010 as an in-memory distributed data grid, and it has evolved into a unified real-time data platform, shaped by customer demand and technology advancement. The GridGain platform combines streaming data in motion and historical data at rest with functionality to help organizations handle complex analytical, streaming, and transactional data workloads at ultra-low latency. In September 2014, GridGain donated its code base to the Apache Software Foundation to create the Apache Ignite open source software project and, by February 2015, it will become a top-level project that provides distributed high-performance computing capabilities. GridGain continues to contribute to the Apache Ignite project, runs an annual Ignite Summit conference, and offers a community edition of its platform based on Apache Ignite. Initially focused on financial services, GridGain has enjoyed expansion into other verticals in recent years, with its high-performance processing of streaming data resulting in the modernization of workloads in new use cases.

Quick facts about GridGain include:

- **Product name:** GridGain Unified Real-Time Data Platform
- **Employees:** 135
- **Global footprint:** Deployed in 31 countries across 3 global regions
- **Top industry areas:** Financial services, telecommunications, transportation, and logistics
- **Cloud:** Available as-a-service on Amazon Web Services, Microsoft Azure, and Google Cloud

- **Pricing model:** Pay as you go or annual subscription
- **Partner ecosystem:** Multiple partners in all major global regions
- **Interesting fact:** GridGain's customers have been able to successfully replace mainframe systems, document databases, and relational databases with the GridGain unified platform, achieving significant performance improvements processing streaming and at-rest data.

GridGain is not always considered for stream processing solutions because of its origins in distributed caching. This foundation of a distributed architecture is what provides GridGain with its ability to handle real-time streaming data and support scalability up to petabytes of in-memory data. GridGain's memory-first, disk-second architecture minimizes disk I/O to provide ultra-low latencies across transactional, streaming, and analytical processing. Co-located compute enables complex processing application logic workloads where the data is.

GridGain does not have any AI-based automation in the current product, but its road map includes plans to introduce AI capabilities, both to leverage AI for improving user experience and for enabling the GridGain platform to serve as the ultra-low-latency data processing platform for AI workloads.

Strengths

- **Deployment, compatibility, and interoperability.** GridGain supports multiple deployment alternatives, including in customer-managed datacenters, on private clouds, and on the public cloud, and is offered as a multitenant software as a service in the public cloud. GridGain supports many different types of event brokers and many different data formats, and it supports the use of the popular application and data engineering coding languages. GridGain supports multiple runtime interfaces, including the JMX API. GridGain supports push and persistent query methods and ultra-real-time latency (less than 40ms).
- **Onboarding and customer success.** GridGain is focused on building initial engagement with customers; understanding key success factors, desired outcomes, and timelines; and aligning them with customer success and services teams to work toward these defined goals. GridGain also conducts periodic architecture reviews and best practices sessions with customers to ensure solution designs and implementations are an optimal use of the platform.

Challenges

- **Proprietary stream processing engine.** GridGain acts as the execution engine for stream processing pipelines and, as such, competes with other open source stream processing engines. GridGain can work in conjunction with other open source stream execution engines and split/share the execution workload, leveraging the Apache Beam programming model.
- **Market presence.** GridGain is still a relatively small fish in a big pond, having a multifaceted solution for distributed data processing, which makes it hard to focus on just one market. GridGain is not available across all regional markets, and its current partner program is limiting growth opportunities in emerging markets. Challenged by hyperscalers and platform providers, GridGain faces significant competitive forces.

Consider GridGain When

Consider GridGain when you seek to work with data in real time, hybrid, or multicloud environments, seamlessly combining data in-motion with data at-rest with compute functionality to handle complex analytical, streaming, and transactional data workloads at ultra-low latency. Alternatively, your organization has made investments in building distributed computing solutions leveraging the Apache Ignite project and is looking for the commercially supported version. Popular use cases of GridGain also include data infrastructure modernization including cloud migration.

Hazelcast

After a thorough evaluation of Hazelcast's strategies and capabilities, IDC has positioned the company in the Major Players category in this IDC MarketScape for worldwide analytic stream processing software.

Hazelcast was founded in 2012 and is headquartered in Palo Alto, California, United States. Hazelcast was born as a distributed in-memory grid in support of building new digital applications across distributed compute resources and in modernizing legacy applications. Hazelcast has transformed itself through a new vision, enabling every business to act instantly and realize real-time potential by putting intuitive and innovative real-time capabilities into the hands of the business. Hazelcast Platform and its as-a-service Hazelcast Cloud solution integrate its stream processing engine, which became generally available in 2019, with a fast data store into a single solution to simplify real-time architectures for next-generation applications and AI/ML deployments. Hazelcast's customers are leveraging the technology to drive new revenue, mitigate risk, and operate efficiently. The Hazelcast code is available on GitHub for use under Apache license version 2.0 and the Hazelcast Community License. Hazelcast Platform is a commercial product that offers additional features to the open source edition (Hazelcast OSS), including the security suite and blue/green deployments for clients.

Quick facts about Hazelcast include:

- **Product name:** Hazelcast Platform and Hazelcast Cloud
- **Employees:** ~175
- **Global footprint:** Operates in all global regions
- **Top industry areas:** Financial services, retail, and high tech
- **Cloud:** Hazelcast Platform available in the AWS, GCP, and Azure marketplaces, and Hazelcast Cloud currently hosted on AWS; support for Hazelcast Cloud on the other two platforms is on the road map
- **Pricing model:** Hazelcast Cloud priced at a per gigabyte per hour model; the customer-managed version — Hazelcast Platform — priced by the number of instances of the software
- **Partner ecosystem:** Global partner ecosystem
- **Interesting fact:** A major bank is leveraging Hazelcast to offer loans to customers in real-time while they are at an ATM. This has resulted in a 4x conversion rate compared with prior batch processing methods that sent email offers one or two days after the deposit was made.

The vision at Hazelcast is to "enable every business to act instantaneously and realize their real-time potential," which requires putting innovative real-time capabilities into the hands of every business. Hazelcast is doing this by offering simplified development and deployment of event stream processing applications through its distributed platform architecture. Hazelcast has also demonstrated performance at a scale of 1 billion events per second in a published benchmark and low-latency advantages in a benchmark from the Hasso Platner Institute at the University of Potsdam. Hazelcast has fostered a vibrant community around its technology, supporting meetups and offering comprehensive training programs.

Hazelcast is building AI capabilities into its platform to improve the developer experience, accelerate customer time to market and value of solutions deployed in Hazelcast, and help customers identify new data-driven opportunities. Hazelcast is also looking to leverage AI to automatically generate data-driven insights, optimize resource allocation, and dynamically adapt to changing workloads across its distributed computing framework.

Strengths

- **Deployment, compatibility, and interoperability.** Hazelcast supports multiple deployment alternatives, including in customer-managed datacenters, private clouds, and the public cloud, and is offered as a multitenant software as a service in the public cloud. Hazelcast Platform supports many different types of event brokers and many different data formats, and it supports the use of the popular application and data engineering coding languages. Hazelcast Platform supports multiple runtime interfaces, including the JMX API. Hazelcast supports all stream processing windowing functions; push, pull, and persistent querying methods; and ultra-real-time latency (less than 40ms), enabling high throughput, resilience, and simplicity.
- **Customer engagement.** Hazelcast has a thorough customer onboarding approach and stays engaged with customers on a regular basis through quarterly calls focused on assuring success, performing health checks, and discussing road map items. The company's customer advisory board provides input in group settings to collaborate on prioritization of road map items. It also has a product strategy board made up of highly technical members in the customer base and invites customers into the Hazelcast innovation lab for co-engineering initiatives.

Challenges

- **Proprietary stream processing engine.** Hazelcast Platform acts as the execution engine for stream processing pipelines and, as such, competes with open source stream processing engines. Hazelcast can work in conjunction with these other streaming execution engines and split/share the execution workload, leveraging the Apache Beam programming model.
- **Market awareness.** Hazelcast's origins as a distributed in-memory data grid popular in data and mainframe modernization use cases doesn't reflect what the technology and the company have evolved into today. Hazelcast is working to overcome this stereotype through its marketing efforts and community building. Hazelcast could also look to expand its technology and channel partnerships to boost growth opportunities in more regions and emerging economies.

Consider Hazelcast When

Consider Hazelcast when you are considering the need for a real-time, resilient, and high-performance data storage, management, and processing solution that can handle analytical, streaming, and transactional workloads at very low latencies across distributed systems. Your data environment is hybrid and/or multicloud, and the availability of industry-specific templates is of importance. Hazelcast also offers license model flexibility if that is a key driver for procurement.

Microsoft

After a thorough evaluation of Microsoft's strategies and capabilities, IDC has positioned the company in the Leaders category in this IDC MarketScape for worldwide analytic stream processing software.

Microsoft was founded in 1975 and is headquartered in Redmond Washington, United States. Azure Stream Analytics first became generally available in 2015. Azure Stream Analytics is a fully managed stream processing engine that is designed to analyze and process large volumes of streaming data with submillisecond latencies. Patterns and relationships can be identified in data that originates from a variety of input sources including applications, devices, sensors, clickstreams, and social media feeds. These patterns can be used to trigger actions and initiate workflows such as creating alerts, feeding information to a reporting tool, or storing transformed data for later use. Stream Analytics is also available on the Azure IoT Edge runtime, enabling to process data directly on IoT devices.

Quick facts about Microsoft include:

- **Product name:** Azure Stream Analytics
- **Employees**: 221,000
- **Global footprint:** Available in all global regions
- **Top industry areas:** Manufacturing, retail, and energy
- **Cloud:** Available on Microsoft Azure
- **Pricing model:** Pay as you go, pay what you use, and by streaming unit hours
- **Partner ecosystem**: Global partner ecosystem
- **Interesting fact:** Microsoft Event Hubs is an Azure ring zero service, which means it is core to the operations of Azure datacenters, and it needs to be available in each region to support Microsoft Azure's business. Azure Streaming Analytics is used by Microsoft to process 14 trillion events every month in billing operations for the Azure business across all regions.

In July of 2023, Microsoft introduced a new version of Azure Stream Analytics at a price point that made it more attractive to customers. Customers are still on a steady migration path to the new product to lower the cost of running stream analytics. Microsoft has been on a rapid pace of innovation with the product, continuously improving it and adding new capabilities. Advanced scenarios support reference data from SQL tables or storage files and, with the integration of Azure ML, real-time scoring is also possible. Microsoft Fabric was announced in 2023, a unified software-as-a-service platform bringing together many of the disparate Microsoft data technologies, and this will include real-time events. Microsoft Fabric event streams will capture, transform, and stream data into fabric destinations for further processing and analytics as needed.

Azure Stream Analytics integrates with Azure Machine Learning to bring AI-driven automation into real-time data stream processing, supporting real-time scoring and predictions, and identify patterns, trends, and relationships within high volumes of data streaming from devices or processes. Microsoft believes generative AI will completely transform streaming analytics by providing users with the ability to express complex analytical questions in plain English.

Strengths

- **Developer experience.** Azure Stream Analytics offers low-code, no-code, code-friendly, and code-only user interfaces for streaming pipeline development. The service provides multiple resiliency features and logging capabilities and promises a high SLA.
- **Market presence.** Microsoft Azure is available across all global regions and has an extensive partner network with a healthy balance of direct and indirect sales.

Challenges

- **Compatibility and interoperability.** Azure Stream Analytics is a proprietary execution engine for stream processing pipelines and only connects to Kafka or Azure Event Hubs event brokers. It primarily supports SQL as a programming language, JavaScript is secondary, and C# is currently in preview. This may be irrelevant, however, because Microsoft also offers a no-code developer experience. It primarily supports CSV, JSON, Avro, and a proprietary event format. It has some support for XML, Thrift, Protobuf, MessagePack, and BSON.
- **Vendor lock-in.** Azure Stream Analytics is available on the Microsoft Azure Cloud, connecting to event brokering technologies implemented in the Microsoft Azure Cloud or to hybrid data sources via secure networking. Microsoft Stream Analytics can be implemented as a

customermanaged service in on-premises or other infrastructure, providing access to additional brokers.

Consider Microsoft When

Consider Microsoft when your organization has a disposition toward Microsoft Azure as a cloud service provider and, as such, much of your streaming data infrastructure is also contained within the Microsoft Azure Cloud. While Azure Stream Analytics can be used for many horizontal and vertical use cases, its most popular use cases include predictive maintenance, real-time fraud detection, and customer behavior prediction.

Oracle

After a thorough evaluation of Oracle's strategies and capabilities, IDC has positioned the company in the Major Players category in this IDC MarketScape for worldwide analytic stream processing software.

Oracle was founded in 1977 and is now headquartered in Austin, Texas, United States. Oracle is a publicly traded company and operates in all global markets. Oracle's event stream processing capability is part of its GoldenGate product portfolio, which is perhaps best known for change data capture and database replication use cases. Change data capture is a commonly used source of streaming data and, therefore, including streaming data processing in the GoldenGate portfolio is a natural fit. Oracle GoldenGate Stream Analytics (GGSA) allows users to process and analyze large-scale real-time information by using sophisticated correlation patterns, enrichment, and machine learning. It offers real-time actionable business insight on streaming data and automates actions that can be used to drive business.

Quick facts about Oracle include:

- **Product name:** GoldenGate Stream Analytics
- **Employees:** 164,000
- **Global footprint:** Available across all global regions
- **Top industry areas:** Financial services, retail, and telecommunications
- **Cloud:** Available on Oracle Cloud Infrastructure, Amazon Web Services, Microsoft Azure, and Google Cloud
- **Pricing model:** Pay as you go (OCPU/hour) as well as subscription and perpetual licenses for on-premises and third-party clouds
- **Partner ecosystem:** Global partner ecosystem
- **Interesting fact:** A competitive sailing race program uses Oracle Stream Analytics for Race Telemetry in real time. Telemetry equipment using wireless access via a 4G network allows race control to push courses, mark positions, and race timer control and conversational chatter to the chart plotter on each support and race boat while also fueling statistics being shown in media broadcasts covering the races.

Oracle GoldenGate Stream Analytics is differentiated in the market as it is anchored in the change data capture origins, where event-based transactional integration is enabling stream data processing, with a breadth and depth of real-time low-impact connectors to most sources of OLTP data, as well as Kafka and JMS/MQ sources. GGSA has a complete self-service web-based graphical user interface design tier for developing streaming pipelines, utilizing a point-and-click no-code framework for specifying data catalog entries, connection information, pipeline development, and lineage information.

GGSA has the capability to build custom visualizations in real-time dashboards and includes geospatial mapping. GGSA also provides built-in patterns for complex event processing (CEP), including event sequencing; geospatial (distance, speed, proximity), financial, trend, and anomaly detection; and identifying missing or duplicate events.

GGSA integrates directly to production machine learning models via industry standard PMML or Open Neural Network Exchange (ONNX) integrations, as well as Oracle ML. Integration with inference engines or third-party rules engines is supported. This integration with Oracle Machine Learning can invoke external text transformers to create and store vector embeddings for further use with Oracle 23.4 vector store, compatible with ONNX large language models, enabling real-time flow of high-quality data into vector embeddings used by modern generative AI applications.

Strengths

- **Monitoring and logging.** GGSA provides detailed logging, notifications and alerts, and stream tracing and lineage, effective for troubleshooting and impact analysis. Operational metrics are available in a dashboard and via API, and monitoring APIs are available. GGSA also supports the full spectrum of time notions in stream processing, including transaction boundary time for events coming from OLTP data sources.
- **Market presence and availability.** GGSA is available across all global regions and fully supported as Oracle works with and continues to invest in its channels and implementation partner programs. GGSA is also available on all major hyperscalers and Oracle Cloud Infrastructure.

Challenges

- **Compatibility and interoperability.** GGSA supports Spark streaming and its own proprietary GoldenGate stream processing engines, but the complexity of these engines is abstracted through the development environment, and therefore, which engine is being used may not be a concern for developers. GGSA natively supports CSV, JSON, Avro, and its own proprietary data formats and, through customization, it can support XML, Thrift, Protobuf, MessagePack, and BSON. GGSA natively supports Kafka, RabbitMQ, and RocketMQ, which are among the most popular brokering technologies and, through JMS, GGSA can support additional event sources and sinks. Java and SQL (CQL dialect) are the programming languages supported by GGSA.
- **Market awareness.** The GoldenGate brand in the market has strength, but it is more likely associated with data replication and not considered for stream processing and analytics, making perception a challenge. It is also important to note that while some look at Oracle and think that its solutions are centered on Oracle databases, GoldenGate originally came through an acquisition and, as such, it supports a wide array of sources and targets, with much of its revenue coming from solutions that include non-Oracle sources and targets.

Consider Oracle When

Consider Oracle when you are an organization that is using transactions within OLTP databases as a primary source of event data used in streaming analytics, and you are looking for a solution that also supports Kafka and other event brokering technologies via JMS. Your organization may already have a relationship with Oracle and may already be using GoldenGate for database replication, perhaps in hybrid and multicloud data environments.

SAS

After a thorough evaluation of SAS' strategies and capabilities, IDC has positioned the company in the Major Players category in this IDC MarketScape for worldwide analytic stream processing software.

SAS Institute Inc., more commonly known as SAS, is a private company headquartered in Cary, North Carolina, United States. SAS was founded by Jim Goodnight and John Sall at North Carolina State University in 1976. SAS has been committed to being at the forefront of data and analytics throughout its history, and SAS is now a multinational software company specializing in advanced analytics, business intelligence, and data management. SAS Event Stream Processing (ESP) was first released in 2013 to leverage the analytic capabilities of SAS alongside the high-performance nature of real-time complex event processing engine. SAS ESP has its roots in the IoT, but it has evolved into a solution that can be applied to any event stream processing use case across multiple industries. SAS ESP's mission is to be the always-on universal inferencing engine in the cloud and at the edge, being the real-time operationalization platform for models produced by a company's data science function.

Quick facts about SAS include:

- **Product name:** SAS Event Stream Processing
- **Employees:** 12,000+
- **Global footprint:** 100 offices worldwide
- **Top industry areas:** Financial services, manufacturing, and telecommunications
- **Cloud:** Available on Amazon Web Services, Microsoft Azure, and Google Cloud
- **Pricing model:** License based on the volume of events processed
- **Partner ecosystem:** Global partner ecosystem
- **Interesting fact:** SAS has installed solar power-generation panels on a 5-acre farm adjacent to its headquarters in Cary, North Carolina The solar farm generates 1.7 million kilowatt hours per year, contributing to carbon dioxide emissions reduction by more than 1,000 tons annually. The field is maintained by a squad of 36 sheep, and SAS ESP is used in the upkeep of the solar assets. SAS ESP is used to proactively monitor for failures in operation as well as predicting the expected power yield based on environmental conditions.

SAS Event Stream Processing offers an integrated, efficient, and reliable approach to event stream processing. It simplifies development, maintenance, and collaboration with built-in analytics, ultimately helping organizations derive insights and take actions from streaming data more effectively and with confidence. This breadth of analytics can result in faster and more informed decision-making in a wide range of applications. SAS Event Stream Processing can connect to diverse data sources; streamline data ingestion, enabling interoperability with existing data ecosystems, adapting to various industries; and offer resilience against evolving data technology.

With the rise of natural language processing, chatbots, and generative AI, SAS' approach revolves around enhancing the efficiency, effectiveness, and ease of working with event streams. SAS envisions a situation where a user would describe the nature of a streaming project to initiate and AI guides the user through the process of configuring data source and schema, adding appropriate processing techniques in a smart wizard-driven interface. There will also be opportunities to use AI for optimization and troubleshooting, suggesting best practices for common misconfigurations, and instances where very complex techniques are being employed.

Strengths

- **Compatibility and deployment.** SAS supports connectivity to multiple types of event brokers, use of multiple message formats, and popular programming languages. As a serverless solution, it can run anywhere, on any cloud, and in hybrid environments. SAS also has broad support for vertical industries with prebuilt templates. SAS leverages its depth of analytics by enabling the use of analytical models in-stream during event processing.
- **Customer onboarding and ongoing engagement.** Commencing with a deep understanding of the customer's unique needs and goals, onboarding includes technical setup, training, proof of concept (POC), use case implementation, continuous monitoring, and responsive customer support. SAS also offers a self-service learning option through the SAS Learning Center, complemented by email and personalized assistance to guide customers. In addition, jointly developed success plans are tailored to expedite value realization for each customer's unique journey. Ongoing customer engagement includes robust support, training, user communities, feedback mechanisms, regular updates, and a consultative approach.

Challenges

- **Proprietary stream processing engine.** By design, SAS utilizes its own proprietary streaming engine, not relying on Spark or Flink, but it does support the use of Kafka Streams in situations where Kafka is being used as the underlying event broker. Use of its own proprietary technology does, however, provide native utilization of analytical models and features of the broader SAS Viya platform. It also enables SAS to operate on low-power edge devices (NVIDIA ORIN hardware, for example) as well as in scalable cloud environments.
- **Competitive forces.** SAS has been a long-tenured incumbent in the advanced data analytics software market, but it has been facing new competition from open source communities and start-up software companies focused on the modern data platform. SAS may be viewed as a legacy solution in the market, but it is increasingly known for the robustness of its streaming analytic capabilities. SAS extends its functional scope through comprehensive integration with key open source frameworks, notably Python and ONNX, empowering users to seamlessly infuse analytics and machine learning methodologies into real-time operational applications. SAS is growing its partner program, focusing on marketing and sales, inclusive of a focus on real-time analytics, with comprehensive support, training, consulting, and demos to help organizations quickly get over learning curves associated with event stream processing.

Consider SAS When

While SAS Event Stream Processing can handle multiple types of events, and cross-industry solutions, its origins are in the Internet of Things. If your company is looking for an IoT analytics solution that can also provide advanced analytics across additional areas of the business and use cases, SAS could be a good choice. In addition, SAS is a good alternative for those looking to a cloud-neutral alternative that can run in hybrid and multicloud environments.

Spotfire

After a thorough evaluation of Spotfire's strategies and capabilities, IDC has positioned the company in the Major Players category in this IDC MarketScape for worldwide analytic stream processing software.

Spotfire is one of the four business units inside Cloud Software Group, a new commercial technology entity created because of the merger between Citrix Systems and TIBCO. The Spotfire business unit has three primary product lines in Spotfire Business Intelligence (BI), Spotfire Streaming, and Spotfire

Data Science. The Spotfire Streaming product is a rebranding of TIBCO StreamBase and is aligned in the Spotfire business under Cloud Software Group, unlocking the ability to have real-time dashboards and analytics in the Spotfire suite. Spotfire Streaming has 150 connectors for ingesting and emitting streaming data, a proprietary EventFlow language designed for stream processing, streaming business intelligence, alerting and actions, and streaming data science capabilities. The Spotfire Streaming development environment is fully integrated into Eclipse. The EventFlow language compiles into byte code, which runs in a Java virtual machine, optimized for cloud-native architectures, but it is not yet available as a service. Spotfire has introduced a new user experience called Data Streams for Cloud, which provides a browser-based interface to connect to event brokering technologies such as Kafka, Pulsar, MQTT, TIBCO eFTL, and looping files.

Quick facts about Spotfire include:

- **Product name:** Spotfire Streaming
- **Employees:** 1,500 (estimate)
- **Global footprint:** Operates in all global markets except China
- **Top industry areas:** Financial services, manufacturing, and energy
- **Cloud:** Can run on cloud infrastructure, but not available as a service
- **Pricing model:** Spotfire Streaming priced per core or per instance
- **Partner ecosystem:** Global partner ecosystem
- **Interesting fact:** In a recent conversation with a large financial services company, our advocate mentioned that it took a large development team over 14 months to deliver a solution built on Kafka Streams that his small team of five developers were able to build in two weeks in streaming.

The EventFlow language has been in use for more than 20 years, and it is the core programming language in Spotfire Streaming. EventFlow is a visual programming language providing a universal way to express handling and processing of data across all supported streams, messaging protocols, FIX venues, and databases. Spotfire Streaming is a complex tool that requires hands-on experience for the developer to understand the value it provides, but once it is mastered, it can become an effective and powerful tool for developers and data engineers, providing productivity gains over hand-coding. The platform is highly extensible such that customers can develop custom operators, adapters, monitor listeners, and so forth to enable self-sufficiency and tailoring for customer-specific scenarios.

The platform supports real-time scoring of AI/ML predictive models from Spotfire Data Science, Python, R (TERR), Spark MLib, H2O, PMML, and TensorFlow. Streaming applications can execute models based on predefined schedules or based on rules/event logic. Models can be updated at runtime without requiring application downtime, and automated alerts communicate with external systems or notify users of models outside of error bounds. Spotfire envisions generative AI unlocking no-code capabilities such as creating streaming applications from textual descriptions of requirements and desired functionality and generating code for front-end interfaces to layer on top of streaming applications. Generative AI is also expected to play a fundamental role in streaming automation in the back-end infrastructure required to connect, integrate, and prepare real-time data for consumption by foundation models and feeding real-time predictions, forecasts, and risk computations from conventional ML models into foundation model results.

Strengths

- **Operations and runtime interfaces.** Spotfire streaming has comprehensive support of monitoring and management of the solution from detailed logging to dashboards and monitoring APIs while also supporting multiple runtime interfaces for DevOps efficiencies. Spotfire streaming also supports multiple types of data formats and event brokering technologies.
- **Customer engagement through community.** Spotfire has built a community around EventFlow and stream processing. Spotfire parallels this community with comprehensive customer and partner support programs.

Challenges

- **Proprietary technology and limited deployment alternatives.** Spotfire utilizes its own proprietary stream processing engine, not leveraging open source projects such as Spark or Flink. Spotfire's objective is to integrate with the standards of the day, providing a consistent platform for handling both legacy and emerging streaming and messaging standards. Spotfire Streaming runs as a macroservice deployable in virtual machines and on bare metal within client-managed environments. This restriction is driven by performance requirements and customer preferences for private controlled environments.
- **Brand transformation.** The formation of Cloud Software Group, the result of the TIBCO and Citrix combination, is taking the assets of both organizations on a different trajectory. Spotfire, Jaspersoft, TIBCO, and Information Builders, all representing TIBCO acquisitions, are being reborn as individual business units under Cloud Software Group, compared with how TIBCO was previously working to integrate capabilities into unified platforms and go-to-market strategies. As a result, Spotfire is focused on transforming its brand, partner program, and market presence into what the business is now and not what it was a year ago, possibly diverting investment away from product research and development toward marketing and sales.

Consider Spotfire When

Consider Spotfire when your organization may not have a cloud-first strategy or at a minimum has a hybrid or multicloud strategy and has sufficient resources to support customer-managed software running in virtual machines or on bare metal. You are looking for a solution that provides ease of development and maintenance in a developer-friendly integrated development environment (IDE), and a primary focus on stream processing is for real-time data visualization and analytics.

Vendors to Watch

Gigaspaces

Gigaspaces' Smart Data Integration Hub (Smart DIH) is an in-memory distributed data store technology that supports high-performance, real-time access to and processing of data at rest and data in motion in event streams. The functional capabilities of the Gigaspaces Smart DIH product are sufficient that it would have qualified for inclusion in this IDC MarketScape if annual revenue was higher and it had a broader geographical presence. Data pipelines can collect data from multiple sources via event-driven or out-of-the-box connectors to a variety of data sources, assembled with no-code configuration, with full ACID transaction support, data validity logic, and the ability to monitor data volume metric anomalies. The Smart DIH provides hot and warm data storage tiering to balance performance with cost, controlling data placement with business logic and support scalability. Organizations that are looking for event brokering, event mesh, and event processing capabilities in

one product that offers high-concurrency, low-latency microservice access to data in-memory should put Gigaspaces on the evaluation list.

IBM

IBM's Event Automation was released in June of 2023, making it ineligible for inclusion in this IDC MarketScape. The IBM Event Automation is intended to put business events to work by enabling users to detect situations, act in real time, and automate decisions to improve business outcomes. At design time, a visual editor is used to connect event sources to sequences of operations, with ease-of-use enabling pipeline development by multiple personas. At run time, IBM provides a fully supported and managed Apache Flink engine for stream processing. When combined with IBM Event Streams, IBM provides an offering complete with the ability to collect streams of real-time events using Apache Kafka, manage event endpoints in a self-service catalog, and define business situations that require automation of real-time decision and action. IBM's event stream processing portfolio of products are available in Kubernetes containers that can run in hybrid cloud environments, supported by IBM and its worldwide network of businesses and partners. As IBM continues to develop and mature its event stream processing capabilities, it will be participating in the market category of software vendors and service providers running managed Apache Flink implementations.

Redpanda

While Redpanda's self-hosted stream processing offering didn't meet the revenue requirement for this MarketScape, it is an event streaming platform intended to provide the infrastructure for real-time data streaming and asynchronous event processing and scalability. Founded in 2019 and headquartered in San Francisco, California, United States, Redpanda's focus is on simplifying event-driven communication and bringing down cost by maximizing utilization of compute power.

APPENDIX

Reading an IDC MarketScape Graph

For the purposes of this analysis, IDC divided potential key measures for success into two primary categories: capabilities and strategies.

Positioning on the y-axis reflects the vendor's current capabilities and menu of services and how well aligned the vendor is to customer needs. The capabilities category focuses on the capabilities of the company and product today, here and now. Under this category, IDC analysts will look at how well a vendor is building/delivering capabilities that enable it to execute its chosen strategy in the market.

Positioning on the x-axis, or strategies axis, indicates how well the vendor's future strategy aligns with what customers will require in three to five years. The strategies category focuses on high-level decisions and underlying assumptions about offerings, customer segments, and business and go-to-market plans for the next three to five years.

The size of the individual vendor markers in the IDC MarketScape represents the market share of each individual vendor within the specific market segment being assessed. For this IDC MarketScape, vendor size was determined using IDC's 2022 Software Tracker, which represents IDC's estimate of each vendor's 2022 software product revenue.

IDC MarketScape Methodology

IDC MarketScape criteria selection, weightings, and vendor scores represent well-researched IDC judgment about the market and specific vendors. IDC analysts tailor the range of standard characteristics by which vendors are measured through structured discussions, surveys, and interviews with market leaders, participants, and end users. Market weightings are based on user interviews, buyer surveys, and the input of IDC experts in each market. IDC analysts base individual vendor scores, and ultimately vendor positions on the IDC MarketScape, on detailed surveys and interviews with the vendors, publicly available information, and end-user experiences in an effort to provide an accurate and consistent assessment of each vendor's characteristics, behavior, and capability.

Market Definition

Analytic stream processing software ingests and processes events by normalizing the events into a data structure, correlating the data to the correct stream, applying time window-based logic to evaluate the data, querying additional data to apply business logic, and may include the use of algorithms to apply advanced analytical models to the event stream and routing or delivering the output to business logic that determines what to do with the output. Stream processing supports a wide range of use cases that support automation and automated decision-making, including descriptive and predictive analytics.

Strategies and Capabilities Criteria

Tables 1 and 2 provide key strategy and capability measures, respectively, for the worldwide analytic stream processing software market.

TABLE 1

Key Strategy Measures for Success: Worldwide Analytic Stream Processing

Strategies Criteria	Definition	Weights (%)
Financial/funding	▪ Length of time in the market and estimated number of customers for the entire company ▪ Revenue for the software vendor in the application development and deployment secondary market as measured and estimated by IDC ▪ IDC's estimated 2022 over 2021 revenue growth ▪ Investments the company is making across its operations and the percentage of current operating budget being invested in research and development (R&D) ▪ Presence in global regions and number of countries	20.0
Partner ecosystem and community	▪ Number of channel, technology, and implementation partners ▪ Developer community and support initiatives available ▪ Extent of availability in the public cloud	15.0
Growth	▪ Ratio of direct-to-indirect sales and extent to which partner programs are being improved	16.0

TABLE 1

Key Strategy Measures for Success: Worldwide Analytic Stream Processing

Strategies Criteria	Definition	Weights (%)
	▪ Estimation of new partners that will be added in 2024 ▪ Geographical distribution of partners	
Momentum	▪ IDC's interpretation of the vendor's perception of differentiations and challenges and level of alignment with how the vendor scored in this evaluation ▪ The vendor's estimate of YoY pipeline growth, combined with the vendor's five year historical revenue CAGR as estimated by IDC	11.0
Innovation	▪ Make-up of research and development team, partner R&D strategy, improvements to R&D, and where AI is being incorporated into the product	6.0
Functionality or offering	▪ Customer visibility of road map and delivery record of road map items ▪ Security and privacy regulations met by the vendor ▪ IDC's interpretation of the vendor's road map and whether or not it is focused on opportunities based on how the vendor scored in this IDC MarketScape	11.0
Customer success	▪ Customer onboarding and continued engagement programs	6.0
Customer delivery strategy	▪ Reference customer satisfaction with the software vendor ▪ Reference customer perception of the vendor's road map ▪ Reference customers' perception about whether or not they feel the vendor has a solution built for the future ▪ Reference customer perception of the level of automation and intelligence within the software technology ▪ Reference customer perception of the software vendor's ability to run in the public cloud	15.0
Total		100.0

Source: IDC, 2024

TABLE 2

Key Capability Measures for Success: Worldwide Analytic Stream Processing

Capabilities Criteria	Definition	Weights (%)
Product	▪ Length of time in market and whether the software is available standalone or as part of a platform ▪ Estimated range of worldwide customers and IDC's estimated product revenue for 2021 and 2022 and YoY growth rates ▪ Customer net retention rate for this product	15.0
Functionality — supported use cases	▪ Business use cases supported by the product ▪ Horizontal use cases supported by the product ▪ Current support of predictive AI use cases ▪ Current support of interpretive AI use cases ▪ Current support of generative AI use cases	12.0
Functionality — monitoring and	▪ Availability and level of logging, alerts and notifications, stream tracing, operational metrics APIs, dashboards, and monitoring APIs	3.0
Functionality — stream processing	▪ Stream processing methods, event handling, windowing, and query methods supported ▪ State management, latencies, and notions of time supported by the product ▪ Data formats supported by the product ▪ Targets supported by the product ▪ Developer experience — availability of different types of developer tools	23.0
Functionality — resiliency	▪ High availability and failover capabilities ▪ Event management in the event of exceptions or failover	6.0
Architecture — security and access	▪ Role-based access, support for OAuth, and availability of audit logs ▪ Event discovery, catalog, schema registry, lineage, and sensitive data tagging capabilities	5.0

TABLE 2

Key Capability Measures for Success: Worldwide Analytic Stream Processing

Capabilities Criteria	Definition	Weights (%)
Architecture — compatibility	▪ Support for various types of deployment methods ▪ Runtime interfaces available ▪ Stream and broker engines supported ▪ Programming languages and frameworks supported ▪ Frequency of releases and extent of backward compatibility ▪ Support for industry templates ▪ Product-specific training availability ▪ Use of AI inside the product	16.0
Pricing model — cost and flexibility	▪ Variability of licensing options available ▪ Basis of how the product is priced ▪ Flexibility of licensing	10.0
Customer satisfaction	▪ Reference customer perception of value for money ▪ Reference customer perception on ease of use ▪ Reference customer perception of performance ▪ Reference customer implementation experience	10.0
Total		100.0

Source: IDC, 2024

LEARN MORE

Related Research

- *IDC MarketScape: Worldwide Event Brokering Software 2024 Vendor Assessment* (IDC #US50301923, March 2024)
- *IDC PeerScape: Practices to Improve Analytic Stream Processing* (IDC #US51712624, March 2024)
- *IDC Market Glance: Data Control Plane, 1Q24* (IDC #US51070924, February 2024)
- *Modern Data Streaming* (IDC #US49529023, August 2023)
- *IDC PlanScape: Future of Intelligence – Data Engineering* (IDC #US47348321, June 2023)

Synopsis

This IDC MarketScape on worldwide analytic stream processing software vendor assessment for 2024 discusses the increasing need for real-time analytic solutions in businesses and growth of the streaming analytics software market. This document evaluates various vendors based on their stream processing capabilities and market strategies. It also provides advice for technology buyers and outlines the vendor inclusion criteria. The document concludes with a summary of each vendor's strengths, challenges, and considerations for potential buyers.

"The speed of business has increased, as organizations need to respond to and make decisions based on what is happening now, not what happened yesterday, last week, or last month, " said Stewart Bond, vice president, Data Intelligence and Integration Software Research at IDC. "Organizations require real-time analytics solutions fed by streams of data; the market has responded and is experiencing significant growth, but not all analytic stream processing products are created equal, thus requiring an objective evaluation of what is in the market."

About IDC

International Data Corporation (IDC) is the premier global provider of market intelligence, advisory services, and events for the information technology, telecommunications, and consumer technology markets. With more than 1,300 analysts worldwide, IDC offers global, regional, and local expertise on technology, IT benchmarking and sourcing, and industry opportunities and trends in over 110 countries. IDC's analysis and insight helps IT professionals, business executives, and the investment community to make fact-based technology decisions and to achieve their key business objectives. Founded in 1964, IDC is a wholly owned subsidiary of International Data Group (IDG, Inc.).

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